

More Examples on Multivariable Derivatives and Preview of Multivariable Integration

Lecture 8 for Multivariable Calculus

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1 Lecture Scope and Goals

This lecture is a bridge lecture. We first consolidate the main derivative tools by working through five representative problems:

- the mean value theorem along line segments,
- constancy results from directional information,

- chain rule calculations with several layers of variables,
- implicit differentiation for first and second partial derivatives,
- a combined problem involving constraints, implicit functions, and differentiation.

After these examples, we begin the transition to multivariable integration.

Central Theme

Differentiation studies local rate of change. The mean value theorem, chain rule, and implicit differentiation are three ways of transporting local derivative information along curves, through compositions, and across equations.

2 A Practical Mean Value Theorem in Several Variables

Theorem 2.1 (Mean Value Theorem along a segment). *Let $D \subset \mathbb{R}^n$ be open, and suppose the line segment from a to b lies in D . If $f : D \rightarrow \mathbb{R}$ is differentiable, then there exists $t_0 \in (0, 1)$ such that*

$$f(b) - f(a) = \nabla f(a + t_0(b - a)) \cdot (b - a).$$

Proof idea. Define a one-variable function

$$\phi(t) = f(a + t(b - a)), \quad 0 \leq t \leq 1.$$

Then the ordinary one-variable mean value theorem gives

$$\phi(1) - \phi(0) = \phi'(t_0).$$

By the chain rule,

$$\phi'(t) = \nabla f(a + t(b - a)) \cdot (b - a).$$

How to Use It

When a problem asks you to estimate $f(P) - f(Q)$, connect P and Q by a line segment and apply the one-variable mean value theorem to f restricted to that segment.

3 Five Representative Problems

3.1 Problem 1: Estimate a Function Difference by the Mean Value Theorem

Let

$$f(x, y) = \ln(1 + x^2 + y^2).$$

Prove that for any two points P, Q in the unit disk

$$\overline{B(0, 1)} = \{(x, y) : x^2 + y^2 \leq 1\},$$

we have

$$|f(P) - f(Q)| \leq \|P - Q\|.$$

Solution. Write $P = (x_1, y_1)$ and $Q = (x_2, y_2)$. Since the closed unit disk is convex, the line segment from P to Q lies in the disk. By the mean value theorem along the segment,

$$f(P) - f(Q) = \nabla f(R) \cdot (P - Q)$$

for some point $R = (x, y)$ on that segment.

Now

$$\nabla f(x, y) = \left(\frac{2x}{1 + x^2 + y^2}, \frac{2y}{1 + x^2 + y^2} \right).$$

Hence

$$\|\nabla f(x, y)\| = \frac{2\sqrt{x^2 + y^2}}{1 + x^2 + y^2}.$$

Let $r = \sqrt{x^2 + y^2}$. On the unit disk $0 \leq r \leq 1$, so

$$\|\nabla f(x, y)\| = \frac{2r}{1 + r^2} \leq 1,$$

because

$$2r \leq 1 + r^2 \iff (r - 1)^2 \geq 0.$$

Therefore

$$|f(P) - f(Q)| = |\nabla f(R) \cdot (P - Q)| \leq \|\nabla f(R)\| \|P - Q\| \leq \|P - Q\|.$$

What this example teaches. The mean value theorem turns a global difference into a derivative estimate. The key is bounding $\|\nabla f\|$ on the region.

3.2 Problem 2: A Constancy Result from a Radial Derivative Condition

Let

$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}.$$

Suppose $f \in C^1(D)$ and

$$xf_x(x, y) + yf_y(x, y) = 0$$

for every $(x, y) \in D$. Prove that f is constant on D .

Solution. Fix an arbitrary point $(x, y) \in D$. Define

$$\phi(t) = f(tx, ty), \quad 0 \leq t \leq 1.$$

For $0 \leq t \leq 1$,

$$(tx)^2 + (ty)^2 = t^2(x^2 + y^2) < 1,$$

so $(tx, ty) \in D$.

For $0 < t \leq 1$, the chain rule gives

$$\phi'(t) = xf_x(tx, ty) + yf_y(tx, ty).$$

Using the given condition at the point (tx, ty) ,

$$txf_x(tx, ty) + tyf_y(tx, ty) = 0.$$

Since $t > 0$, divide by t :

$$xf_x(tx, ty) + yf_y(tx, ty) = 0.$$

Thus

$$\phi'(t) = 0, \quad 0 < t \leq 1.$$

So ϕ is constant on $[0, 1]$. Therefore

$$f(x, y) = \phi(1) = \phi(0) = f(0, 0).$$

Since (x, y) was arbitrary, f is constant on D .

What this example teaches. The expression

$$xf_x + yf_y$$

is the derivative of f in the radial direction. If this derivative is always zero, then f does not change along any ray from the origin.

3.3 Problem 3: A Chain Rule Computation with Two Layers

Let

$$z = f(u, v), \quad u = x^2 - y^2, \quad v = 2xy,$$

where f has continuous second partial derivatives. Express

$$z_{xx} + z_{yy}$$

in terms of derivatives of f with respect to u and v .

Solution. First compute the first partial derivatives:

$$u_x = 2x, \quad u_y = -2y, \quad v_x = 2y, \quad v_y = 2x.$$

By the chain rule,

$$z_x = f_u u_x + f_v v_x = 2xf_u + 2yf_v,$$

and

$$z_y = f_u u_y + f_v v_y = -2yf_u + 2xf_v.$$

Here f_u, f_v are evaluated at (u, v) .

Differentiate z_x with respect to x :

$$z_{xx} = 2f_u + 2x(f_{uu}u_x + f_{uv}v_x) + 2y(f_{vu}u_x + f_{vv}v_x).$$

Using $u_x = 2x$ and $v_x = 2y$,

$$z_{xx} = 2f_u + 4x^2 f_{uu} + 4xy f_{uv} + 4xy f_{vu} + 4y^2 f_{vv}.$$

Since $f_{uv} = f_{vu}$,

$$z_{xx} = 2f_u + 4x^2 f_{uu} + 8xy f_{uv} + 4y^2 f_{vv}.$$

Differentiate z_y with respect to y :

$$z_{yy} = -2f_u - 2y(f_{uu}u_y + f_{uv}v_y) + 2x(f_{vu}u_y + f_{vv}v_y).$$

Using $u_y = -2y$ and $v_y = 2x$,

$$z_{yy} = -2f_u + 4y^2 f_{uu} - 4xy f_{uv} - 4xy f_{vu} + 4x^2 f_{vv}.$$

Thus

$$z_{yy} = -2f_u + 4y^2 f_{uu} - 8xy f_{uv} + 4x^2 f_{vv}.$$

Adding the two expressions,

$$z_{xx} + z_{yy} = 4(x^2 + y^2)f_{uu} + 4(x^2 + y^2)f_{vv}.$$

Therefore

$$\boxed{z_{xx} + z_{yy} = 4(x^2 + y^2)(f_{uu} + f_{vv})}.$$

What this example teaches. Second derivatives under a change of variables are longer than first derivatives. Keep track of which variables each derivative is taken with respect to.

3.4 Problem 4: Implicit Differentiation and Second Partial Derivatives

Suppose $z = z(x, y)$ is determined near $(0, 0)$ by

$$z + xe^z + y^2 - 1 = 0,$$

with

$$z(0, 0) = 1.$$

Compute

$$z_x(0, 0), \quad z_y(0, 0), \quad z_{xx}(0, 0).$$

Solution. Let

$$F(x, y, z) = z + xe^z + y^2 - 1.$$

Then

$$F_z = 1 + xe^z.$$

At $(0, 0, 1)$,

$$F_z(0, 0, 1) = 1 \neq 0.$$

So the implicit function theorem applies locally.

Differentiate

$$F(x, y, z(x, y)) = 0$$

with respect to x :

$$F_x + F_z z_x = 0.$$

Now

$$F_x = e^z, \quad F_z = 1 + xe^z.$$

Thus

$$z_x = -\frac{e^z}{1 + xe^z}.$$

At $(0, 0)$, where $z = 1$,

$$\boxed{z_x(0, 0) = -e}.$$

Differentiate with respect to y :

$$F_y + F_z z_y = 0.$$

Since

$$F_y = 2y,$$

we have

$$z_y = -\frac{2y}{1 + xe^z}.$$

Therefore

$$\boxed{z_y(0, 0) = 0}.$$

To find $z_{xx}(0, 0)$, differentiate

$$F_x + F_z z_x = 0$$

again with respect to x along the implicit function. The standard second derivative formula is

$$F_{xx} + 2F_{xz}z_x + F_{zz}z_x^2 + F_z z_{xx} = 0.$$

Compute

$$F_{xx} = 0, \quad F_{xz} = e^z, \quad F_{zz} = xe^z.$$

At $(0, 0, 1)$,

$$F_{xx} = 0, \quad F_{xz} = e, \quad F_{zz} = 0, \quad F_z = 1, \quad z_x = -e.$$

Hence

$$0 + 2e(-e) + 0 + z_{xx} = 0,$$

so

$$\boxed{z_{xx}(0, 0) = 2e^2}.$$

What this example teaches. For second implicit derivatives, it is cleaner to differentiate $F_x + F_z z_x = 0$ than to differentiate an already complicated explicit formula.

3.5 Problem 5: A Combined Problem with Constraint and Implicit Dependence

Let $z = z(x, y)$ be defined implicitly near $(1, 0)$ by

$$xz + ye^z + x^2 - 1 = 0,$$

and suppose

$$z(1, 0) = 0.$$

Consider the curve on the unit circle

$$x(t) = \cos t, \quad y(t) = \sin t.$$

Compute

$$\left. \frac{d}{dt} z(x(t), y(t)) \right|_{t=0}.$$

Then explain why this derivative can be interpreted as a directional derivative of the implicit function $z(x, y)$ along the tangent direction to the unit circle at $(1, 0)$.

Solution. Let

$$F(x, y, z) = xz + ye^z + x^2 - 1.$$

Then

$$F_x = z + 2x, \quad F_y = e^z, \quad F_z = x + ye^z.$$

At $(1, 0, 0)$,

$$F_z = 1 \neq 0.$$

Thus the implicit function theorem applies locally.

Implicit differentiation gives

$$z_x = -\frac{F_x}{F_z} = -\frac{z + 2x}{x + ye^z}, \quad z_y = -\frac{F_y}{F_z} = -\frac{e^z}{x + ye^z}.$$

At $(1, 0, 0)$,

$$z_x(1, 0) = -2, \quad z_y(1, 0) = -1.$$

Along the curve $(x(t), y(t)) = (\cos t, \sin t)$,

$$\frac{dz}{dt} = z_x x'(t) + z_y y'(t).$$

At $t = 0$,

$$x'(0) = -\sin 0 = 0, \quad y'(0) = \cos 0 = 1.$$

Thus

$$\left. \frac{dz}{dt} \right|_{t=0} = z_x(1, 0) \cdot 0 + z_y(1, 0) \cdot 1 = -1.$$

The tangent vector to the unit circle at $(1, 0)$ is $(0, 1)$, so this derivative is the directional derivative of z in that tangent direction.

What this example teaches. This example combines three ideas: the implicit function theorem gives $z = z(x, y)$, implicit differentiation gives ∇z , and the chain rule evaluates the derivative along a constrained curve.

4 Transition to Multivariable Integration

4.1 From Local Rate to Accumulated Total

So far, derivatives answered questions such as:

- How fast does f change near a point?
- What is the best linear approximation?
- How does a constraint affect possible motion?
- When can an equation determine one variable as a function of others?

Multivariable integration asks a complementary question:

If a quantity is spread over a region, what is the total amount?

4.2 High-Dimensional Integration and Measure Elements

The idea of integration is always the same:

Break a region into very small pieces, multiply the value of the function by the size of each piece, and add everything up.

What changes from one dimension to higher dimensions is the meaning of “size.”

One dimension: length. On an interval $[a, b] \subset \mathbb{R}$, the small piece is a short interval of length Δx .

The limiting measure element is written

$$dx.$$

Thus

$$\int_a^b f(x) \, dx$$

adds values of f weighted by length.

Two dimensions: area. In the plane, the small piece is usually a small rectangle with side lengths

$$\Delta x, \quad \Delta y.$$

Its area is

$$\Delta A = \Delta x \, \Delta y.$$

The limiting area element is written

$$dA,$$

and in Cartesian coordinates,

$$dA = dx \, dy \quad \text{or} \quad dA = dy \, dx.$$

Thus

$$\iint_D f(x, y) \, dA$$

adds values of f weighted by area.

Three dimensions: volume. In space, the small piece is a small rectangular box with side lengths

$$\Delta x, \quad \Delta y, \quad \Delta z.$$

Its volume is

$$\Delta V = \Delta x \, \Delta y \, \Delta z.$$

The limiting volume element is written

$$dV,$$

and in Cartesian coordinates,

$$dV = dx \, dy \, dz$$

with the order chosen according to the iterated integral. Thus

$$\iiint_E f(x, y, z) \, dV$$

adds values of f weighted by volume.

General dimension. In $\mathbb{R}^n$, the small box has side lengths

$$\Delta x_1, \Delta x_2, \dots, \Delta x_n,$$

and its n -dimensional volume is

$$\Delta V_n = \Delta x_1 \Delta x_2 \cdots \Delta x_n.$$

The corresponding measure element is often written

$$dV_n \quad \text{or} \quad dx_1 dx_2 \cdots dx_n.$$

For a region $E \subset \mathbb{R}^n$,

$$\int_E f \, dV_n$$

means the total accumulation of f over E with respect to n -dimensional volume.

Notation Summary

Dimension	Measure element	Meaning
1	dx	length
2	dA	area
3	dV	volume
n	dV_n	n -dimensional volume

Integrability Fact Used in This Textbook

In this course, we will use the following standard fact:

If f is continuous on a closed and bounded region E , then f is integrable on E .

In particular, the integral is finite:

$$\int_E f \, dV_n$$

exists as a finite number. For plane regions this means

$$\iint_D f(x, y) \, dA$$

is finite, and for solid regions this means

$$\iiint_E f(x, y, z) \, dV$$

is finite.

4.3 The Basic Double Integral

If $f(x, y)$ represents density over a plane region D , then

$$\iint_D f(x, y) \, dA$$

represents the total mass over D .

For a rectangle

$$R = [a, b] \times [c, d],$$

the double integral is defined as the limit of Riemann sums:

$$\iint_R f(x, y) \, dA = \lim_{\max \Delta A_{ij} \rightarrow 0} \sum_{i,j} f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}.$$

Exercise: compute an integral from the definition. Let

$$D = \{(x, y) : x^2 + y^2 \leq 1\}.$$

Use the definition of the double integral to compute

$$\iint_D x^2 \, dA.$$

Solution. We approximate the disk by small squares in the plane. Put the disk inside the square

$$[-1, 1] \times [-1, 1].$$

Divide this square into small squares of side length

$$h = \frac{2}{N}.$$

For each vertical strip, choose a sample point x_i^* in the interval

$$x_{i-1} \leq x \leq x_i.$$

The cross-section of the disk above this value of x has height

$$2\sqrt{1 - (x_i^*)^2}.$$

Using small squares, the total area of the squares in this vertical strip is approximately

$$2\sqrt{1 - (x_i^*)^2} \Delta x,$$

where $\Delta x = h$.

On this vertical strip, the value of the integrand is approximately

$$(x_i^*)^2.$$

Thus the Riemann sum using small squares is approximately

$$\sum_i (x_i^*)^2 \cdot 2\sqrt{1 - (x_i^*)^2} \Delta x.$$

As the side length of the squares tends to 0, this Riemann sum converges to

$$\iint_D x^2 \, dA = \int_{-1}^1 2x^2 \sqrt{1 - x^2} \, dx.$$

Now compute this one-variable integral. Since the integrand is even,

$$\int_{-1}^1 2x^2 \sqrt{1-x^2} \, dx = 4 \int_0^1 x^2 \sqrt{1-x^2} \, dx.$$

Let

$$x = \sin t, \quad 0 \leq t \leq \frac{\pi}{2}.$$

Then

$$dx = \cos t \, dt, \quad \sqrt{1-x^2} = \cos t.$$

Therefore

$$4 \int_0^1 x^2 \sqrt{1-x^2} \, dx = 4 \int_0^{\pi/2} \sin^2 t \cos^2 t \, dt.$$

Using

$$\sin^2 t \cos^2 t = \frac{1}{4} \sin^2(2t),$$

we get

$$4 \int_0^{\pi/2} \sin^2 t \cos^2 t \, dt = \int_0^{\pi/2} \sin^2(2t) \, dt = \frac{\pi}{4}.$$

Hence

$$\boxed{\iint_D x^2 \, dA = \frac{\pi}{4}}.$$

Geometric interpretation: double integral as volume. The small-square approximation also explains the geometry of a double integral. Over each small square ΔA_{ij} in the region D , the surface

$$z = f(x, y)$$

has height approximately

$$f(x_{ij}^*, y_{ij}^*).$$

So the small column standing above that square has volume approximately

$$f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}.$$

Adding all these small column volumes gives the Riemann sum

$$\sum_{i,j} f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}.$$

Taking the limit gives

$$\iint_D f(x, y) \, dA.$$

Therefore, when $f(x, y) \geq 0$ on D ,

$$\boxed{\iint_D f(x, y) \, dA = \text{the volume under the surface } z = f(x, y) \text{ and above the region } D.}$$

For the present exercise, $\iint_D x^2 \, dA$ is the volume of the solid above the unit disk and below the surface

$$z = x^2.$$

Remark. This agrees with symmetry:

$$\iint_D x^2 \, dA = \iint_D y^2 \, dA$$

and therefore

$$2 \iint_D x^2 \, dA = \iint_D (x^2 + y^2) \, dA = \int_0^{2\pi} \int_0^1 r^2 \cdot r \, dr \, d\theta = \frac{\pi}{2}.$$

4.4 First Computation Principle: Iterated Integrals

For many functions on rectangles,

$$\iint_R f(x, y) \, dA = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy.$$

Example. For

$$R = [0, 1] \times [0, 2], \quad f(x, y) = x + y,$$

we compute

$$\iint_R (x + y) \, dA = \int_0^1 \int_0^2 (x + y) \, dy \, dx.$$

The inner integral is

$$\int_0^2 (x + y) \, dy = 2x + 2.$$

Thus

$$\int_0^1 (2x + 2) \, dx = [x^2 + 2x]_0^1 = 3.$$

4.5 Regions Beyond Rectangles

Many regions are not rectangles. For example, a type I region has the form

$$D = \{(x, y) : a \leq x \leq b, \, g_1(x) \leq y \leq g_2(x)\}.$$

Then

$$\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$

A type II region has the form

$$D = \{(x, y) : c \leq y \leq d, \, h_1(y) \leq x \leq h_2(y)\}.$$

Then

$$\iint_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy.$$

4.6 Changing the Order of Integration

Changing the order of integration means describing the same region in a different way.

Example. Rewrite the integral

$$I = \int_0^1 \int_{x^2}^{\sqrt{x}} f(x, y) \, dy \, dx$$

by reversing the order of integration.

Solution. The original limits describe the region

$$D = \{(x, y) : 0 \leq x \leq 1, \, x^2 \leq y \leq \sqrt{x}\}.$$

The boundary curves are

$$y = x^2, \quad y = \sqrt{x}.$$

Equivalently,

$$x = \sqrt{y}, \quad x = y^2.$$

For $0 \leq y \leq 1$, the horizontal slice runs from

$$x = y^2 \quad \text{to} \quad x = \sqrt{y}.$$

Therefore

$$\int_0^1 \int_{x^2}^{\sqrt{x}} f(x, y) \, dy \, dx = \int_0^1 \int_{y^2}^{\sqrt{y}} f(x, y) \, dx \, dy.$$

A concrete computation. For example,

$$\int_0^1 \int_{x^2}^{\sqrt{x}} x \, dy \, dx = \int_0^1 \int_{y^2}^{\sqrt{y}} x \, dx \, dy.$$

Using the reversed order,

$$\int_0^1 \left[\frac{x^2}{2} \right]_{y^2}^{\sqrt{y}} dy = \frac{1}{2} \int_0^1 (y - y^4) \, dy = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{5} \right) = \frac{3}{20}.$$

Changing Order Checklist

1. Draw or mentally identify the region D .
2. Find the outer variable range in the new order.
3. Describe a typical slice in the new direction.
4. Rewrite the inner limits from left to right or bottom to top.

4.7 Polar Coordinates for Double Integrals

Polar coordinates are useful when the region or integrand has circular symmetry:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad dA = r \, dr \, d\theta.$$

Double integral example. Compute

$$\iint_D (x^2 + y^2) \, dA,$$

where D is the disk

$$D = \{(x, y) : x^2 + y^2 \leq 4\}.$$

Solution. In polar coordinates,

$$x^2 + y^2 = r^2, \quad D = \{(r, \theta) : 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi\}.$$

Thus

$$\iint_D (x^2 + y^2) \, dA = \int_0^{2\pi} \int_0^2 r^2 \cdot r \, dr \, d\theta.$$

So

$$\int_0^{2\pi} \int_0^2 r^3 \, dr \, d\theta = \int_0^{2\pi} \left[\frac{r^4}{4} \right]_0^2 d\theta = \int_0^{2\pi} 4 \, d\theta = 8\pi.$$

Another common polar region. If D is the upper half of the unit disk, then

$$0 \leq r \leq 1, \quad 0 \leq \theta \leq \pi.$$

If D is the sector between angles α and β and radius R , then

$$0 \leq r \leq R, \quad \alpha \leq \theta \leq \beta.$$

4.8 Triple Integrals

If $f(x, y, z)$ represents density in a solid region $E \subset \mathbb{R}^3$, then

$$\iiint_E f(x, y, z) \, dV$$

represents total mass over E .

For a rectangular box

$$B = [a, b] \times [c, d] \times [p, q],$$

we compute by iterated integrals:

$$\iiint_B f(x, y, z) \, dV = \int_a^b \int_c^d \int_p^q f(x, y, z) \, dz \, dy \, dx,$$

with other orders also possible when the function and region are sufficiently nice.

General solid regions. For a solid described by

$$E = \{(x, y, z) : (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\},$$

we write

$$\iiint_E f(x, y, z) \, dV = \iint_D \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \, dA.$$

The main task is usually to describe the projection D and the lower and upper surfaces.

4.9 Cylindrical Coordinates for Triple Integrals

Cylindrical coordinates are polar coordinates in the xy -plane, together with the vertical variable z :

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z, \quad dV = r \, dz \, dr \, d\theta.$$

Triple integral example. Compute

$$\iiint_E (x^2 + y^2) \, dV,$$

where E is the solid cylinder

$$E = \{(x, y, z) : x^2 + y^2 \leq 1, 0 \leq z \leq 2\}.$$

Solution. In cylindrical coordinates,

$$x^2 + y^2 = r^2,$$

and the solid is described by

$$0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq 2.$$

Therefore

$$\iiint_E (x^2 + y^2) \, dV = \int_0^{2\pi} \int_0^1 \int_0^2 r^2 \cdot r \, dz \, dr \, d\theta.$$

Compute:

$$\int_0^{2\pi} \int_0^1 \int_0^2 r^3 \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^1 2r^3 \, dr \, d\theta = \int_0^{2\pi} \frac{1}{2} \, d\theta = \pi.$$

Spherical coordinates preview. For ball-shaped regions, spherical coordinates are often better:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

with

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

For example, the volume of the ball $\rho \leq R$ is

$$\int_0^{2\pi} \int_0^\pi \int_0^R \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{4}{3}\pi R^3.$$

Spherical coordinate example. Compute

$$\iiint_\Omega y^2 \, dV,$$

where

$$\Omega = \{(x, y, z) : x^2 + y^2 + z^2 \leq 2z\}.$$

Solution. First rewrite the region:

$$x^2 + y^2 + z^2 \leq 2z \iff x^2 + y^2 + (z - 1)^2 \leq 1.$$

Thus Ω is the ball of radius 1 centered at $(0, 0, 1)$.

Using spherical coordinates centered at the origin,

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

The inequality becomes

$$\rho^2 \leq 2\rho \cos \phi.$$

Since $\rho \geq 0$, this gives

$$0 \leq \rho \leq 2 \cos \phi.$$

We also need $\cos \phi \geq 0$, so

$$0 \leq \phi \leq \frac{\pi}{2}, \quad 0 \leq \theta \leq 2\pi.$$

Now

$$y^2 = \rho^2 \sin^2 \phi \sin^2 \theta, \quad dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

Therefore

$$\iiint_{\Omega} y^2 \, dV = \int_0^{2\pi} \int_0^{\pi/2} \int_0^{2 \cos \phi} \rho^4 \sin^3 \phi \sin^2 \theta \, d\rho \, d\phi \, d\theta.$$

Separate the variables:

$$\iiint_{\Omega} y^2 \, dV = \left(\int_0^{2\pi} \sin^2 \theta \, d\theta \right) \left(\int_0^{\pi/2} \sin^3 \phi \int_0^{2 \cos \phi} \rho^4 \, d\rho \, d\phi \right).$$

We have

$$\int_0^{2\pi} \sin^2 \theta \, d\theta = \pi,$$

and

$$\int_0^{2 \cos \phi} \rho^4 \, d\rho = \frac{(2 \cos \phi)^5}{5} = \frac{32}{5} \cos^5 \phi.$$

Thus

$$\iiint_{\Omega} y^2 \, dV = \pi \cdot \frac{32}{5} \int_0^{\pi/2} \sin^3 \phi \cos^5 \phi \, d\phi.$$

Using the standard beta integral, or a direct substitution, one gets

$$\int_0^{\pi/2} \sin^3 \phi \cos^5 \phi \, d\phi = \frac{1}{24}.$$

Hence

$$\boxed{\iiint_{\Omega} y^2 \, dV = \frac{4\pi}{15}}.$$

Second method: project to the xy -plane first. We can also compute the same integral by describing Ω using vertical slices. From

$$x^2 + y^2 + (z - 1)^2 \leq 1,$$

the projection onto the xy -plane is

$$D = \{(x, y) : x^2 + y^2 \leq 1\}.$$

For each fixed $(x, y) \in D$, the variable z satisfies

$$(z - 1)^2 \leq 1 - x^2 - y^2.$$

Hence

$$1 - \sqrt{1 - x^2 - y^2} \leq z \leq 1 + \sqrt{1 - x^2 - y^2}.$$

Therefore

$$\iiint_{\Omega} y^2 \, dV = \iint_D \int_{1-\sqrt{1-x^2-y^2}}^{1+\sqrt{1-x^2-y^2}} y^2 \, dz \, dA.$$

Since the integrand does not depend on z , the inner integral is simply the height of the slice:

$$\int_{1-\sqrt{1-x^2-y^2}}^{1+\sqrt{1-x^2-y^2}} y^2 \, dz = 2y^2 \sqrt{1 - x^2 - y^2}.$$

Thus

$$\iiint_{\Omega} y^2 \, dV = \iint_D 2y^2 \sqrt{1 - x^2 - y^2} \, dA.$$

Now use polar coordinates in the xy -plane:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad dA = r \, dr \, d\theta.$$

Then

$$0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi,$$

and

$$\iiint_{\Omega} y^2 \, dV = \int_0^{2\pi} \int_0^1 2r^2 \sin^2 \theta \sqrt{1 - r^2} r \, dr \, d\theta.$$

Separate the variables:

$$\iiint_{\Omega} y^2 \, dV = 2 \left(\int_0^{2\pi} \sin^2 \theta \, d\theta \right) \left(\int_0^1 r^3 \sqrt{1 - r^2} \, dr \right).$$

The angular integral is

$$\int_0^{2\pi} \sin^2 \theta \, d\theta = \pi.$$

For the radial integral, let $u = 1 - r^2$. Then $r \, dr = -\frac{1}{2} du$ and $r^2 = 1 - u$, so

$$\int_0^1 r^3 \sqrt{1 - r^2} \, dr = \frac{1}{2} \int_0^1 (1 - u) u^{1/2} \, du = \frac{1}{2} \left(\frac{2}{3} - \frac{2}{5} \right) = \frac{2}{15}.$$

Therefore

$$\boxed{\iiint_{\Omega} y^2 \, dV = 2 \cdot \pi \cdot \frac{2}{15} = \frac{4\pi}{15}}.$$

A symmetry shortcut. Since Ω is the ball of radius 1 centered at $(0, 0, 1)$, translate $w = z - 1$. Then the region becomes

$$x^2 + y^2 + w^2 \leq 1.$$

By symmetry over the unit ball,

$$\iiint_{\Omega} y^2 \, dV = \frac{1}{3} \iiint_{x^2+y^2+w^2 \leq 1} (x^2 + y^2 + w^2) \, dV.$$

In spherical coordinates around the center,

$$\frac{1}{3} \int_0^{2\pi} \int_0^{\pi} \int_0^1 \rho^2 \cdot \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{1}{3} \cdot \frac{4\pi}{5} = \frac{4\pi}{15}.$$

4.10 General Change of Variables Formula and the Jacobian

The coordinate changes above are special cases of the general change of variables formula.

Two variables. Suppose

$$x = x(u, v), \quad y = y(u, v),$$

and the transformation maps a region S in the uv -plane to a region D in the xy -plane. Then

$$\boxed{\iint_D f(x, y) \, dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, du \, dv}.$$

Here

$$\frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix} = x_u y_v - x_v y_u$$

is the Jacobian determinant.

Polar coordinates as an example. For

$$x = r \cos \theta, \quad y = r \sin \theta,$$

we have

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r.$$

Therefore

$$dA = r \, dr \, d\theta.$$

Three variables. Suppose

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w).$$

Then

$$\boxed{\iiint_E f(x, y, z) \, dV = \iiint_G f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| \, du \, dv \, dw}.$$

The Jacobian determinant is

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \det \begin{pmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{pmatrix}.$$

Geometric meaning. The absolute value of the Jacobian measures local area or volume scaling. In one variable, a substitution uses the factor $|dx/du|$. In several variables, the Jacobian determinant is the corresponding scaling factor.

4.11 Additional Example: The Gaussian Integral

This example shows how a one-variable integral with no elementary antiderivative can be evaluated by turning its square into a double integral.

Problem. The function e^{-x^2} has no elementary antiderivative, so the goal is not to find an elementary primitive. Instead, let $a > 0$ and define

$$I(a) = \int_0^a e^{-x^2} dx,$$

and

$$J(a) = \iint_{D_a} e^{-x^2-y^2} dx dy,$$

where

$$D_a = \{(x, y) : x^2 + y^2 \leq a^2, x \geq 0, y \geq 0\}.$$

Prove that

$$\lim_{a \rightarrow +\infty} \int_0^a e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

Step 1: Write the square as a double integral. Since

$$I(a) = \int_0^a e^{-x^2} dx,$$

we have

$$[I(a)]^2 = \left(\int_0^a e^{-x^2} dx \right) \left(\int_0^a e^{-y^2} dy \right).$$

By Fubini's theorem,

$$\boxed{[I(a)]^2 = \int_0^a \int_0^a e^{-x^2-y^2} dy dx = \iint_{R_a} e^{-x^2-y^2} dA},$$

where

$$R_a = \{(x, y) : 0 \leq x \leq a, 0 \leq y \leq a\}.$$

Step 2: Compare regions. The quarter disk D_a is contained in the square R_a . Also, the square R_a is contained in the quarter disk of radius $\sqrt{2}a$:

$$D_a \subset R_a \subset D_{\sqrt{2}a}.$$

Since the integrand $e^{-x^2-y^2}$ is positive, we obtain

$$\boxed{J(a) \leq [I(a)]^2 \leq J(\sqrt{2}a)}.$$

Step 3: Compute $J(a)$ in polar coordinates. On the quarter disk,

$$0 \leq r \leq a, \quad 0 \leq \theta \leq \frac{\pi}{2}.$$

Since

$$x^2 + y^2 = r^2, \quad dA = r \, dr \, d\theta,$$

we get

$$J(a) = \int_0^{\pi/2} \int_0^a e^{-r^2} r \, dr \, d\theta.$$

The inner integral is

$$\int_0^a e^{-r^2} r \, dr = \left[-\frac{1}{2} e^{-r^2} \right]_0^a = \frac{1}{2} (1 - e^{-a^2}).$$

Therefore

$$J(a) = \frac{\pi}{2} \cdot \frac{1}{2} (1 - e^{-a^2}) = \frac{\pi}{4} (1 - e^{-a^2}).$$

Taking $a \rightarrow +\infty$ gives

$$\lim_{a \rightarrow +\infty} J(a) = \frac{\pi}{4}, \quad \lim_{a \rightarrow +\infty} J(\sqrt{2}a) = \frac{\pi}{4}.$$

By the squeeze theorem,

$$\lim_{a \rightarrow +\infty} [I(a)]^2 = \frac{\pi}{4}.$$

Since $I(a) > 0$, we conclude that

$$\boxed{\lim_{a \rightarrow +\infty} \int_0^a e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}}.$$

Equivalently,

$$\boxed{\int_0^{+\infty} e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}}.$$

What this example teaches. Some one-variable integrals become computable only after being lifted to higher dimensions. The key tools are product integrals, comparison of regions, and polar coordinates.

4.12 Companion Example: Replacing x^2 by x^4

Now replace x^2 by x^4 and consider

$$K = \int_0^{+\infty} e^{-x^4} \, dx.$$

Again, e^{-x^4} has no elementary antiderivative. However, the computation is different from the Gaussian case.

Step 1: Squaring still gives a double integral. For $a > 0$, define

$$K(a) = \int_0^a e^{-x^4} dx.$$

Then

$$[K(a)]^2 = \left(\int_0^a e^{-x^4} dx \right) \left(\int_0^a e^{-y^4} dy \right) = \int_0^a \int_0^a e^{-x^4-y^4} dy dx.$$

Thus

$$\boxed{[K(a)]^2 = \iint_{R_a} e^{-x^4-y^4} dA}, \quad R_a = \{(x, y) : 0 \leq x \leq a, 0 \leq y \leq a\}.$$

Why ordinary polar coordinates no longer simplify it. In polar coordinates,

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Then

$$x^4 + y^4 = r^4(\cos^4 \theta + \sin^4 \theta),$$

so the integrand becomes

$$e^{-r^4(\cos^4 \theta + \sin^4 \theta)}.$$

This does not separate into a purely radial part times a purely angular constant in the same simple way as

$$e^{-x^2-y^2} = e^{-r^2}.$$

Therefore the Gaussian polar-coordinate trick is special to the power 2.

Step 2: Compute the one-variable integral using the Gamma function. Recall the Gamma function:

$$\Gamma(s) = \int_0^{+\infty} t^{s-1} e^{-t} dt, \quad s > 0.$$

For

$$K = \int_0^{+\infty} e^{-x^4} dx,$$

use the substitution

$$t = x^4.$$

Then

$$x = t^{1/4}, \quad dx = \frac{1}{4} t^{-3/4} dt.$$

Therefore

$$K = \int_0^{+\infty} e^{-t} \frac{1}{4} t^{-3/4} dt = \frac{1}{4} \int_0^{+\infty} t^{1/4-1} e^{-t} dt.$$

Hence

$$\boxed{\int_0^{+\infty} e^{-x^4} dx = \frac{1}{4} \Gamma\left(\frac{1}{4}\right)}.$$

General pattern. More generally, for $p > 0$,

$$\int_0^{+\infty} e^{-x^p} dx = \frac{1}{p} \Gamma\left(\frac{1}{p}\right).$$

Indeed, the substitution $t = x^p$ gives

$$dx = \frac{1}{p} t^{1/p-1} dt.$$

What this example teaches. Changing x^2 to x^4 keeps the integral non-elementary, but it changes the best method. The Gaussian case is special because $x^2 + y^2$ is exactly the radial quantity in polar coordinates. For x^4 , the Gamma function is the natural language.

Preview for Next Lecture

Next time we will focus on drawing regions, setting up double and triple integrals, changing the order of integration, and choosing Cartesian, polar, cylindrical, or spherical coordinates according to the geometry.

5 Takeaways

1. The multivariable mean value theorem is usually the one-variable mean value theorem applied to a line segment.
2. The chain rule is most reliable when we write down all variable dependences before differentiating.
3. Implicit differentiation is legal only after checking the relevant nonzero partial derivative condition.
4. Directional and constrained derivatives connect naturally to curves.
5. Double and triple integrals convert local density into total accumulation over a region or solid.
6. Changing coordinates changes the area or volume element: $dA = r dr d\theta$ in polar coordinates and $dV = r dz dr d\theta$ in cylindrical coordinates.
7. The general coordinate-change factor is the absolute value of the Jacobian determinant.